Temple University
Department of Economics

Econ 8007 Macroeconomic Theory I

Fall 2026

Instructor: Pedro Silos
Office: Gladfelter Hall 218

Course meetings: MW 12:30pm–1:45pm
Credit Hours: 3

Contact information:
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You must use your Temple email for all communications. During the week I will normally respond within 24 hours. If I have not responded within 48 hours, please send a follow-up email.
Office Hours: MW 2:00pm–3:00pm, T 1:00pm-2:00pm and by appointment. See me during office hours for any problem related to our course or to your completion of any assignment for the course.

Student Support Services

The following academic support services are available to support you:

1. Student Success Center
2. University Libraries
3. Undergraduate Research Support
4. Career Center
5. Tuttleman Counseling Services
6. Disability Resources and Services

If you are experiencing food insecurity or financial struggles, Temple provides resources and support. Notably, the Temple University Cherry Pantry and the Temple University Emergency Student Aid Program are in operation as well as resources from the Office of Student Affairs.

Prerequisites

The prerequisite of this course is ECON8002. Essentials: dynamic programming under uncertainty, the income fluctuation problem, and stationary equilibrium in incomplete-markets economies (Huggett/Aiyagari). Familiarity with a scientific computing language (Julia, Python, MATLAB, or similar) is assumed; the assignments require you to solve models and to work with some microdata.

Course Objectives

The first weeks of Econ 8007 apply the tools from the first-year sequence to labor markets, inequality, and fiscal policy. There are two goals here. First, to develop two classes of models that ar throughout modern macroeconomics: frictional labor markets (search and matching) and heterogeneous-agent economies with endogenous labor supply and capital accumulation. Second, to connect these frameworks to data. The goal is that by the end of the course you can read and understand current quantitative research.

At the beginning, we will look at some datasets (CPS, JOLTS, and administrative data) and what they tell us about worker and job flows. We then develop the McCall and Diamond–Mortensen–Pissarides (DMP) frameworks and confront them with business-cycle evidence (the Shimer puzzle and its proposed resolutions), followed by life-cycle earnings dynamics and human capital, building directly on the heterogeneity tools from last semester. The second part of the course studies the interaction of capital and labor in heterogeneous-agent economies (endogenous labor supply and aggregation, capital-skill complementarity, and precautionary savings inside frictional labor markets) and closes with two weeks on fiscal policy, studying government debt and transition dynamics, capital taxation, and the progressivity of the tax-and-transfer system.

Readings

There is no required textbook. Lecture notes and papers will carry the course. We will make heavy use of several chapters of *Macroeconomics* (Azzimonti, Krusell, McKay, and Mukoyama), the free textbook for first-year macro PhD courses that we used in 8002—in particular Chapter 20 (Frictional Labor Markets, by Mukoyama and Şahin), Chapter 12 (Labor Supply, by Rogerson and Violante), Chapter 15 (Government and Public Policies, by Azzimonti, Heathcote, and Storesletten), and Chapter 21 (Inequality, by Krusell and Ríos-Rull). Other useful references (do not purchase anything unless you plan to do research in this area; some of these are free and available online):

1. Lars Ljungqvist and Thomas Sargent: *Recursive Macroeconomic Theory*, 4th edition, 2018,

MIT Press. Chapters 6, 29, 30, and 31 cover search, matching, and aggregate labor supply; Chapters 10, 11, 16, 17, and 18 cover fiscal policy and incomplete markets.

2. Christopher Pissarides: *Equilibrium Unemployment Theory*, 2nd edition, 2000, MIT Press.

3. Richard Rogerson, Robert Shimer, and Randall Wright (2005), "Search-Theoretic Models of the Labor Market: A Survey," *Journal of Economic Literature*.

4. QuantEcon Lectures, by Perla, Sargent, and Stachurski (for computation of the models discussed).

Grading

There are four assignments, one replication project, and one final exam. The assignments mix analytical questions with computation and data work (CPS worker flows, calibrating and solving the DMP model, solving an Aiyagari economy with endogenous labor supply, and computing the transition dynamics of a fiscal reform). The replication/extension project asks you to reproduce the main quantitative result of one paper from a list distributed in Week 2. You must suggest an extension and given an oral presentation of the replication strategy with a short (5–6 page) report accompanying it. The final exam is scheduled for December 9th.

Your grade for the first part of 8007:

- Final Exam: 50%
- Assignments: 30%
- Replication/Extension Project: 20%

Course Policies

Expectations for Class Conduct

It is important to foster a respectful and productive learning environment that includes all students in our diverse community of learners. Our differences, some of which are outlined in the University's nondiscrimination statement, add richness to the learning experience. All opinions and experiences, no matter how different or controversial, must be respected in the tolerant spirit of academic discourse. Treat classmates and the instructor with respect in all communication and class activities. Comment, question, or critique ideas—but do not attack individuals. Remember that sarcasm, humor, and slang can be misconstrued; avoid profanity and USING ALL CAPS (which can be perceived as shouting). Protect your own and others' privacy.

Missed Examinations

There will be no make-up tests. If you miss a test, it will be scored as a zero unless you provide written evidence documenting an emergency or acute serious medical condition.

Religious Holidays

If you will be observing any religious holidays that fall on the day of an exam/assignment, I will offer a make-up opportunity. You must inform me of the dates at least a week in advance.

Extra Credit

There will be no extra-credit opportunities. The final grade will be computed according to the formula stated above.

Regrading

If you feel a question was graded unfairly, you may request a regrade by providing a short written statement explaining why you deserve a higher grade. If I accept your request, I will regrade the entire test—your mark may increase or decrease.

Academic Violations

I follow a policy of zero tolerance for violations of academic integrity. Any infractions will receive severe penalties and be reported to the University Disciplinary Committee. See the Student Code of Conduct here and the statement on Student and Faculty Academic Rights and Responsibilities here.

Generative AI

The use of generative AI tools (such as Claude, Gemini, ChatGPT, etc.) is not permitted for the direct completion of tasks assigned in this course. Any use of AI tools in this class with that goal may be considered a violation of Temple University's Academic Honesty policy and Student Conduct Code. However, the use of AI as an alternative source of material or for clarifying concepts explained in class is certainly allowed.

Use of Laptops/Phones in Class

You may not use laptops or phones during class (phones allowed for emergencies). Evidence suggests learning outcomes are worse with laptops for note-taking.

Students with Disabilities

Any student who has a need for accommodations based on the impact of a documented disability or medical condition should contact Disability Resources and Services (DRS) located in the Howard Gittis Student Center South, 4th Floor at drs@temple.edu or 215-204-1280 to request accommodations and learn more about the resources available to you. If you have a DRS accommodation letter to share with me, or you would like to discuss your accommodations, please contact me as soon as practical. I will work with you and with DRS to coordinate reasonable accommodations for all students with documented disabilities. All discussions related to your accommodations will be confidential.

Statement on Academic Freedom

Freedom to teach and freedom to learn are inseparable facets of academic freedom. The University has adopted a policy on Student and Faculty Academic Rights and Responsibilities (Policy #03.70.02) which can be accessed here.

Course Outline

1. **Labor Market Facts: Stocks, Flows, and Measurement.** (*Macroeconomics*, Ch. 20, opening sections on facts; L&S, Section 6.4)

   - Davis, Faberman & Haltiwanger (2006), "The Flow Approach to Labor Markets," *JEP*.
   - Shimer (2012), "Reassessing the Ins and Outs of Unemployment," *Review of Economic Dynamics*.
   - Elsby, Michaels & Ratner (2015), "The Beveridge Curve: A Survey," *JEL*.

2. **Search and Matching: McCall and DMP.** Sequential search; reservation wages; matching functions; Nash bargaining; equilibrium unemployment and the Hosios condition. (*Macroeconomics*, Ch. 20; L&S, Ch. 6, Sections 6.1–6.4, and Ch. 29, Sections 29.2–29.4)

   - Rogerson, Shimer & Wright (2005), Sections 1–4.
   - Pissarides (2000), Chapter 1.
   - QuantEcon "McCall Search Model" and "Lake Model" lectures (for computation).

3. **Unemployment over the Business Cycle: The Shimer Puzzle.** (L&S, Ch. 30, Sections 30.2–30.3; *Macroeconomics*, Ch. 20, quantitative sections)

   - Shimer (2005), "The Cyclical Behavior of Equilibrium Unemployment and Vacancies," *AER*.
   - Hagedorn & Manovskii (2008), "The Cyclical Behavior of Equilibrium Unemployment and Vacancies Revisited," *AER*.
   - Hall (2005), "Employment Fluctuations with Equilibrium Wage Stickiness," *AER*.
   - Ljungqvist & Sargent (2017), "The Fundamental Surplus," *AER*.

4. **Earnings Dynamics and Human Capital over the Life Cycle.** (*Macroeconomics*, Ch. 21; L&S, Section 31.6, Ben-Porath human capital)

   - Ben-Porath (1967), "The Production of Human Capital and the Life Cycle of Earnings," *JPE*.
   - Huggett, Ventura & Yaron (2011), "Sources of Lifetime Inequality," *AER*.
   - Guvenen, Karahan, Ozkan & Song (2021), "What Do Data on Millions of U.S. Workers Reveal About Lifecycle Earnings Dynamics?" *Econometrica*.

5. **Capital and Labor in Heterogeneous-Agent Economies.** (*Macroeconomics*, Ch. 12; L&S, Section 17.8, Section 18.15, and Sections 31.8–31.10)

   - Pijoan-Mas (2006), "Precautionary Savings or Working Longer Hours?" *Review of Economic Dynamics*.

- Chang & Kim (2007), "Heterogeneity and Aggregation: Implications for Labor-Market Fluctuations," *AER*.
- Krusell, Ohanian, Ríos-Rull & Violante (2000), "Capital-Skill Complementarity and Inequality: A Macroeconomic Analysis," *Econometrica*.
- Krusell, Mukoyama & Şahin (2010), "Labour-Market Matching with Precautionary Savings and Aggregate Fluctuations," *Review of Economic Studies*.

6. **Fiscal Policy with Heterogeneous Agents I: Government Debt and Transitions.** Debt and precautionary savings; crowding out in incomplete markets; computing transition dynamics after a policy reform. (*Macroeconomics*, Ch. 15; L&S, Ch. 10, Sections 11.6 and 11.9 for the shooting algorithm and transition experiments, and Section 18.2)

   - Aiyagari & McGrattan (1998), "The Optimum Quantity of Debt," *JME*.
   - Domeij & Heathcote (2004), "On the Distributional Effects of Reducing Capital Taxes," *IER* (for transition dynamics).

7. **Fiscal Policy with Heterogeneous Agents II: Taxation and Labor Supply.** Optimal capital taxation with life-cycle heterogeneity; tax progressivity; taxes and aggregate hours. (*Macroeconomics*, Ch. 15 and Ch. 12, sections on taxation; L&S, Ch. 16, Sections 16.1–16.7, and Sections 31.3–31.5)

   - Conesa, Kitao & Krueger (2009), "Taxing Capital? Not a Bad Idea After All!" *AER*.
   - Heathcote, Storesletten & Violante (2017), "Optimal Tax Progressivity: An Analytical Framework," *QJE*.
   - Prescott (2004), "Why Do Americans Work So Much More Than Europeans?" *FRB Minneapolis Quarterly Review*.